COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	09/18/26	**DATE SUBMITTED:**	09/18/26
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ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

main.cpp > main()

```cpp
#include <iostream>
#include "BST.h"

int main()
{
    // Create the root node
    BinarySearchTree<int> tree(20);

    tree.insert(10);
    tree.insert(5);
    tree.insert(70);
    tree.insert(80);
    tree.insert(40);
    tree.insert(50);

  std::cout << "preOrder: ";
    tree.inOrder();
    std::cout << std::endl;

    std::cout << "inOrder: ";
    tree.inOrder();
    std::cout << std::endl;

    return 0;
}
```

BST.h > BinarySearchTree<T> > preOrder()

```
#ifndef BST_H
#define BST_H

#include <iostream>

template <typename T>
class BinarySearchTree
{
private:
    T key;

    // Pointers
    BinarySearchTree<T>* left;
    BinarySearchTree<T>* right;

public:
    // Constructor
    BinarySearchTree(T value);

    // Insert function
    void insert(T value);

    // Traversing algorithm
    void preOrder ();
    void inOrder();

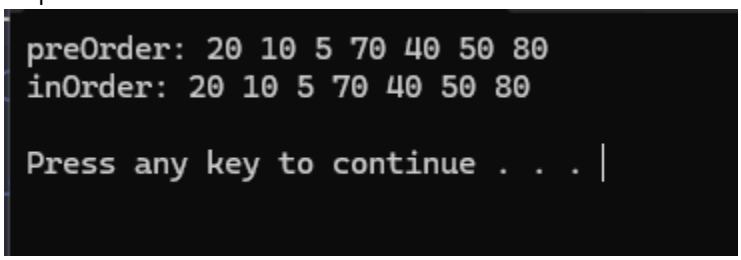
};

// Constructor
template <typename T>
BinarySearchTree<T>::BinarySearchTree(T value)
{
    key = value;
    left = nullptr;
    right = nullptr;
}

// Insert function
template <typename T>
void BinarySearchTree<T>::insert(T value)
{
    if (value < key)
    {
        if (left == nullptr)
        {
            left = new BinarySearchTree<T>(value);
        }
        else
```

```
49              {
50                  left->insert(value);
51              }
52          }
53          else if (value > key)
54          {
55              if (right == nullptr)
56              {
57                  right = new BinarySearchTree<T>(value);
58              }
59              else
60              {
61                  right->insert(value);
62              }
63          }
64      }
65
66      // Pre-order traversal
67      template <typename T>
68      void BinarySearchTree<T>::inOrder()
69      {
70          std::cout << key << " ";
71
72          if (left != nullptr)
73          {
74              left->inOrder();
75          }
76
77          if (right != nullptr)
78          {
79              right->inOrder();
80          }
81      }
82
83      #endif
```

Output:



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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